

Question	Answer	Marks
4(a)	when (two or more) waves meet (at a point)	B1
	(resultant) displacement is the sum of the individual displacements	B1
4(b)(i)	constant phase difference (between the waves)	B1
4(b)(ii)	1. phase difference = 360° or 0	B1
	2. path difference = 1.5λ = 1.5×610 = 920 nm	A1
4(b)(iii)	$\lambda = ax/D$	C1
	$x = 22/4$ (= 5.5 mm) or $22 \times 10^{-3}/4$ (= 5.5×10^{-3} m)	C1
	$a = (610 \times 10^{-9} \times 2.7)/(5.5 \times 10^{-3})$ = 3.0×10^{-4} m	A1
4(b)(iv)	shorter wavelength and (so) separation decreases	B1
4(b)(v)	- no change to fringe separation/fringe width/number of fringes - bright fringes are brighter - dark fringes are unchanged <i>Any two of the above three points, 1 mark each.</i>	B2

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5(c)	region (of space) where a force acts on a (stationary) charge	D1